

# Kartikeya Gangwar

Undergraduate Researcher / Scientific Machine Learning, Geometric DL & Computational Mathematics

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## Research Profile

Undergraduate mathematical researcher investigating the intersection of **Scientific Machine Learning (SciML)**, **Symplectic Geometry**, **Inverse Problems**, and **High-Dimensional Stochastic PDEs**. Author of **10 research manuscripts** (including 4 journal submissions under review: CNAME, TCI, PoF, AMM) and **2 open-source scientific computing packages**. Architect of the “**Adaptive Subspace (AS) Paradigm**”—a cohesive three-act theoretical lineage resolving destructive gradient interference across parameter manifolds, progressing from static algebraic null-space direct-sum decomposition (Act I: *null-space-pinn*), to autonomous parameter-space Adaptive Mesh Refinement in physical PDEs (Act II: *as-pinn*), to dynamic Mixture-of-Experts in transformer latent feature manifolds (Act III: *as-vit-multitask*). Primary methodology emphasizes **structural geometric invariants** (exact symplectic 2-form conservation, Arnold extended contact phase spaces, and B-spline sensitivity projections) over black-box heuristics.

## Education

2023–2027 **B.Sc. (Honours) Mathematics**, *University of Delhi*, Delhi, India

- **Degree Structure:** 4-Year Undergraduate Research Degree with advanced research specialization.
- **Undergraduate Dissertation Supervisor:** Prof. Vinay Kumar (Department of Mathematics).
- **Relevant Core Coursework:** Real Analysis, Multivariable Calculus, Ordinary & Partial Differential Equations, Complex Analysis, Abstract Algebra, Linear Algebra, Numerical Analysis, Probability & Statistics.
- **Independent Scholarly Trajectory:** Rigorous, self-directed specialization in geometric deep learning, symplectic mechanics, nonlinear PDEs, and high-performance scientific computing developed alongside the formal undergraduate program.

## Research Manuscripts & Preprints

Pillar I: Inverse Problems, Symplectic Astrodynamics & Fluid Mechanics

Under Review **Shape Inversion in Electrical Impedance Tomography**, *IEEE Transactions on Computational Imaging (TCI)*

“Derivative-Informed Neural Forward Surrogates with Directional Sensitivity Supervision for Shape Inversion in Electrical Impedance Tomography”

- **Authors:** Kartikeya Gangwar (Sole Author / Lead Investigator).
- **Mathematical Formulation:** Parameterized closed inclusion boundaries via 64-dimensional periodic cubic B-splines coupled to mollified boundary-integral winding indicators. Verified analytic sensitivities  $\mathbf{J}_{\text{FEM}} = \mathbf{J}_{\sigma} \frac{\partial \Sigma}{\partial \theta}$  against directional finite differences ( $< 3.2 \times 10^{-4}$  discrepancy).
- **Algorithmic Novelty:** Introduced **Stochastic Directional JVP Supervision** on  $\mathbb{S}^{63}$ , reducing retained tangent autograd branches from 64 to 1 per sample and bounding peak VRAM to 342.8 MB.
- **Empirical Results:** Zero-online-FEM inversion delivers a **56.8×** wall-clock speedup (35.6 ms vs 2.02 s/step) with a consensus IoU of 0.8235 and a 99.57% noise robustness rate across 234 trials.
- **Codebase:** <https://github.com/KartikeyaGangwar/eit-neural-surrogate-inversion>  
**Preprint DOI:** 10.5281/zenodo.22096368

Working Paper **CPA-SHNN: Symplectic Neural Networks for Celestial Mechanics, Target: Astronomy & Computing / New Astronomy**

*"Causality-Preserving Adaptive Symplectic Hamiltonian Neural Networks (CPA-SHNN) for Autonomous and Non-Autonomous Multi-Body Celestial Gravitational Dynamics"*

- **Authors:** Kartikeya Gangwar (Lead Author), **Prof. Vinay Kumar** (Supervisor & Senior Author).
- **Mathematical Formulation:** Proved *Theorem 1* (Separable Symplectic Kinetic-Coriolis Decomposition,  $\nabla_{\mathbf{z}} \cdot \mathbf{f}_{\theta} \equiv 0$ ) and *Theorem 2* (Arnold Extended Contact Phase Space  $\mathbf{Z}_{\text{ext}} = (\mathbf{q}, t, \mathbf{p}, p_t)$ ,  $\mathcal{K}_{\theta} \equiv 0$ ).
- **Empirical Benchmarks:** Benchmarked across 6 chaotic multi-body celestial systems from Prof. Vinay Kumar's research portfolio (Binary Quasars, Restricted 6-Body, Sitnikov 5-Body, Magnetic Yukawa, Variable-Mass Binaries), preserving energy to machine precision ( $\Delta\mathcal{H} \leq 10^{-4}\%$ ).
- **Ablation Discoveries:** Multi-scale Fourier features achieved up to a **126.40×** error collapse on gravitational saddle singularities. Poincaré generating function ablations demonstrated the necessity of continuous vector fields.
- **Codebase:** <https://github.com/KartikeyaGangwar/cpa-shnn>

Under Review **Operator Conditioning & False Convergence in PINNs, Physics of Fluids (AIP Publishing)**  
*"Operator Conditioning and False Convergence in Physics-Informed Neural Networks for Incompressible Flows"* [MS #POF26-AR-14856]

- **Authors:** Kartikeya Gangwar (Sole Author / Lead Investigator).
- **Formulation Investigation:** Conducted controlled formulation study on 2D regularized lid-driven cavity ( $Re = 1000$ ), contrasting Streamfunction–Pressure ( $\psi$ – $p$ ) against Streamfunction–Vorticity ( $\psi$ – $\omega$ ) across 20,000 Adam + L-BFGS iterations.
- **False Convergence Discovery:** Discovered that  $\psi$ – $\omega$  PINNs suffer from **False Convergence and Operator Diffusion** ( $67.3\times$  residual drop) while underpredicting secondary corner eddy intensity by 14.3% ( $\psi_{\min} = 0.845 \times 10^{-3}$ ) and yielding 14.2% global velocity  $L_2$  error vs 2.8% in  $\psi$ – $p$ .
- **Operator Conditioning & Hodge Projection:** Proved continuous mesh-free PINNs lack discrete stencils for Thom's wall-vorticity condition, leaving boundary vorticity under-specified; retaining pressure  $p$  in  $\psi$ – $p$  enforces an implicit elliptic Helmholtz–Hodge projection that restores operator conditioning.
- **Codebase & Preprint:** <https://github.com/KartikeyaGangwar/pinn-fluid-formulations> **Preprint DOI:** 10.5281/zenodo.22979276

Under Review **Extreme- $Re$  Navier–Stokes Framework & Unsteady Dynamics, Applied Mathematical Modelling (Elsevier)**

*"A High-Resolution Finite-Difference Framework for Steady Continuation and Finite-Window Unsteady Dynamics up to  $Re = 100,000$  in 2D Lid-Driven Cavity Flows"*

- **Authors:** Kartikeya Gangwar (Sole Author / Principal Investigator).
- **Exact Spectral Poisson Acceleration:** Formulated an exact  $\mathcal{O}(N^2 \log N)$  Discrete Sine Transform (DST-I) Poisson engine with vectorized ADI marching, eliminating iterative residual drift and enabling machine-precision pressure-velocity coupling across  $1.05 \times 10^6$  nodes ( $1025 \times 1025$ ).
- **High- $Re$  Frontiers & Batchelor Verification:** Advanced steady continuation up to  $Re = 100,000$  on uniform meshes up to  $1025^2$ , observing core vorticity stabilization ( $\omega_0 \approx -2.54$ ,  $x_c \rightarrow 0.50000$ ) aligned with Batchelor's core theorem; verified numerical uncertainty via ASME V&V 20 Richardson GCI ( $GCI_{32} = 0.48\%$ ).
- **Directional Hybrid Regularization:** Maintained strictly zero artificial viscosity ( $\nu_{\text{art}} = 0$ ) for all  $Re \leq 50,000$  via pure second-order central differencing; regularized corner singularities at  $Re = 100,000$  ( $Pe = 195.3$ ) via directional hybrid differencing with explicit numerical viscosity tensor mapping.
- **Finite-Window Unsteady Marching:** Executed time-accurate marching on  $513 \times 513$  meshes ( $t = 16.0$ – $22.0$  c.t.u.,  $St \equiv fL/U$ ), resolving coherent vortex shedding ( $Re = 10,000$ ,  $St = 0.6249 \pm 0.0416$ ), amplitude-modulated dynamics ( $Re = 15,000$ ,  $St = 0.5364$ ), and finite-window spectral responses up to  $Re = 100,000$ .
- **Open-Source Benchmarks & Datasets:** Released 30 verified flow-field datasets with SHA-256 integrity on Zenodo and GitHub: <https://github.com/KartikeyaGangwar/lid-driven-cavity-cfd> **DOI:** 10.5281/zenodo.18312938

Pillar II: The Adaptive Subspace (AS) Optimization Paradigm

- Under Review **Parameter Decoupling, Conditioning & Interface Transmission**, *Computer Methods in Applied Mechanics and Engineering (CMAME, Elsevier)*  
*"On Parameter Decoupling, Conditioning, and Interface Transmission in Multi-Objective Physics-Informed Neural Networks"* [MS #CMAME-D-26-05120]  
  - **Authors:** Kartikeya Gangwar (First & Corresponding Author), **Samarjeet Malik** (Co-Author).
  - **Mathematical Formulation:** Constructed an algebraic direct-sum decomposition ( $\Theta = \Theta_0 \oplus \Theta_1, \mathcal{W}_0 \mathcal{W}_1^T = 0$ ) with  $C^2$  **Quintic Hermite routing** ( $\psi(\xi) = 6\xi^5 - 15\xi^4 + 10\xi^3$ ), proving exact gradient orthogonality ( $\langle \nabla \mathcal{L}_{\text{if}}, \nabla \mathcal{L}_{\text{des}} \rangle \equiv 0$ ) on saturated domains with zero jump discontinuities ( $\nabla^2 u \in C^0$ ).
  - **Theoretical & Duality Discoveries:** Proved representation rank invariance ( $\text{rank}(\mathbf{H}_\alpha \mathbf{H}_\alpha^T) = 2, \forall \alpha < 1$ ) establishing continuous sharing as an optimizer preconditioner. Using Green's representation theory, proved that compact cubic B-spline bases resolve boundary stiffness ( $1.53\times$  lower interior error,  $2.08 \times 10^{-4}$  interface loss) without parameter splitting.
  - **Multi-Physics Verification:** Evaluated across 10 matched-capacity models ( $P = 9,216$ , 8 seeds) and 6 physical regimes, including 3D Linear Paul Ion Traps ( $2.0\times$  harmonic purity), high-contrast iron magnets, and non-linear phase-field/shock systems.
  - **Codebase & Preprint:** <https://github.com/KartikeyaGangwar/null-space-pinn>  
**Preprint DOI:** 10.5281/zenodo.22132799

Preprint **Adaptive  $N$ -Subspace PINN (AS-PINN)**, *Target: Journal of Computational Physics (JCP) / SISC*  
*"Adaptive  $N$ -Subspace Physics-Informed Neural Networks (AS-PINN): Autonomous Domain Decomposition via Parameter-Space Adaptive Mesh Refinement"*  
  - **Authors:** Kartikeya Gangwar (Sole Author / Principal Investigator).
  - **Algorithmic Novelty:** Developed an autonomous parameter-space AMR engine using vectorized per-sample Gram alignment profiling ( $\mathcal{G}_{ij} = \cos \angle(\mathbf{g}_i, \mathbf{g}_j)$  via `torch.func.vmap`) with exact zero-disruption cleavage invariance ( $\|u^{(N+1)} - u^{(N)}\| = 0$ ).
  - **Empirical Verification:** Validated across 9 canonical PDEs (Burgers, Convection 2D, Quantum NLS, 4D Diffusion, Bratu 2D, Kovasznay, Cavity NS, Klein-Gordon), achieving a **725.6 $\times$**  loss reduction over standard PINNs, PCGrad, and CAGrad on high-frequency Helmholtz ( $k = 4\pi$ ).
  - **Codebase & Preprint:** <https://github.com/KartikeyaGangwar/as-pinn> **Preprint DOI:** 10.5281/zenodo.22822521

Preprint **AS-ViT: Adaptive Subspace Vision Transformers**, *Target: IEEE TPAMI / CVPR*  
*"AS-ViT: Adaptive Subspace Vision Transformers with Autonomous Gradient-Clash Routing for Multi-Task Learning"*  
  - **Authors:** Kartikeya Gangwar (Sole Author / Principal Investigator).
  - **Architectural Innovation:** Resolved negative transfer in dense multi-task vision backbones (segmentation, depth, normals, edges) by continuously tracking inter-task Gram matrix eigenvalues ( $\lambda_{\min}(\mathcal{G}) < -\tau_{\text{conflict}}$ ) and cleaving dedicated latent expert subspaces using Partition of Unity (PoU) gating.
  - **Empirical Results:** Outperformed monolithic ViTs, PCGrad, CAGrad, and fixed MoEs on NYUv2, Cityscapes, and Taskonomy benchmarks, achieving +5.84% mean multi-task gain ( $\Delta M$ ) with zero manual expert tuning.
  - **Codebase & Preprint:** <https://github.com/KartikeyaGangwar/as-vit-multitask>  
**Preprint DOI:** 10.5281/zenodo.22974130

**Pillar III: High-Dimensional Stochastic Systems & Uncertainty Quantification**

Working Paper **High-Dimensional American Basket Options (Deep-EEP-PINN)**, *Target: Quantitative Finance*  
*"Deep Early Exercise Premium Physics-Informed Neural Network for High-Dimensional Free-Boundary Obstacle Problems"*  
  - **Authors:** Kartikeya Gangwar (Lead Author / Principal Investigator).
  - **Mathematical Formulation:** Embedded the analytical Kim (1990) / Carr-Jarrow-Myneni (1992) Early Exercise Premium decomposition into a hard-constrained boundary ansatz:  $V_{\text{Amer}} = V_{\text{Euro}}^{\text{exact}} + e_\theta(\mathbf{S}, t)$  with  $e_\theta(\mathbf{S}, T) \equiv 0$ .
  - **High-Dimensional Breakthrough:** Priced correlated American basket options up to **d = 50 assets (1,225 correlations)**, breaking Bellman's Curse of Dimensionality ( $N^d$ ) via Directional Autograd Hessian Trace Contraction in  $\mathcal{O}(d)$  linear complexity ( $< 3$  GB VRAM). Validated against 100,000-path Longstaff-Schwartz Monte Carlo and Crank-Nicolson PSOR (0.37% relative error).
  - **Codebase:** <https://github.com/KartikeyaGangwar/Deep-EEP-PINN>

- Working Paper **Bayesian PINN Fidelity & Posterior Distortion**, *Target: Reliability Engineering & System Safety / Inverse Problems*  
*"Bayesian PINN Fidelity & Parameter-Resolved Error Localization Framework"*
- **Authors:** Kartikeya Gangwar (Lead Author / Principal Investigator).
  - **Epistemological Investigation:** Formally investigated how PINN forward surrogate approximation errors distort Bayesian inverse posteriors.
  - **Proposed Diagnostic Metric:** Formulated the **Bayesian Fidelity Ratio (BFR)**:  $\text{BFR} = \mathcal{W}_1(\pi_E, \pi_A) / \mathcal{W}_1(\pi_{E,1}, \pi_{E,2})$  based on 1-Wasserstein optimal transport distance normalized by empirical MCMC stochastic noise floors under paired control chains.
  - **Codebase:** <https://github.com/KartikeyaGangwar/PINN-Bayesian-Posterior-Fidelity>

## Scientific Software & Computational Infrastructure

### Pillar IV: Sim2Real Autonomous Vehicle Dynamics & Scientific Infrastructure

- Preprint / v2.0.1 **BeamNG Research SDK (BRSDK)**, *Target: JOSS (2027) / Open-Source*  
*"BRSDK: A High-Frequency In-Engine Telemetry Framework and Python SDK for Sim2Real Autonomous Vehicle Research in BeamNG"*
- **Architect & Lead Developer:** Kartikeya Gangwar (Sole Author / Lead Architect).
  - **Sim2Real Physics Instrumentation:** Formulated an in-engine extraction architecture operating in BeamNG's **2,000 Hz soft-body physics thread** to resolve sub-millisecond transient dynamics (tire slip breakaway energy, dynamic damper velocities, and chassis compliance) necessary for robust simulation-to-real transfer.
  - **Zero-Allocation Memory Design:** Engineered a dual-runtime pipeline eliminating dynamic Lua heap allocations during logging, preserving deterministic simulation stepping and continuous ground-truth physical state trajectories without garbage collection pauses.
  - **Scientific Python SDK & ML Interface:** Built an Apache Arrow columnar Python SDK (brsdk) powered by Polars, featuring strict Pydantic V2 metadata sidecar boundaries and zero-copy DLPack interchange with PyTorch and Minari for offline reinforcement learning.
  - **Codebase & Dissemination:** <https://github.com/KartikeyaGangwar/BRSDK> **DOI:** 10.5281/zenodo.21729606 **Release:** v2.0.1
- Open-Source **Extreme-Re Navier–Stokes Suite & Benchmark Engine**, *Python / NumPy / SciPy*
- **Lead Developer:** Kartikeya Gangwar.
  - **Spectral Poisson Engine:** Integrated an exact  $\mathcal{O}(N^2 \log N)$  Discrete Sine Transform (DST-I) solver, eliminating iterative Poisson residuals and enabling machine-precision pressure-velocity coupling across  $1.05 \times 10^6$  nodes ( $1025 \times 1025$ ).
  - **Continuation & Regularization:** Engineered continuation across 23 Reynolds numbers ( $Re = 100 \rightarrow 100,000$ ), preserving pure central differencing ( $\nu_{\text{art}} = 0$ ) for  $Re \leq 50,000$  and applying directional hybrid differencing at  $Re = 100,000$ .
  - **Verification & Open Dissemination:** Fully automated ASME V&V 20 Richardson GCI verification, real-time unsteady probe telemetry, and animated shedding pipelines; released 30 verified flow-field datasets.
  - **Open-Source Dissemination:** <https://github.com/KartikeyaGangwar/lid-driven-cavity-cfd> **DOI:** 10.5281/zenodo.18312938

## Technical Expertise & Methodologies

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|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scientific Machine Learning          | Physics-Informed Neural Networks (PINNs), Hamiltonian Neural Networks (HNNs), Symplectic Neural ODEs, Deep Surrogate Inversion, Boundary-Encoded Neural Ansätze, Fourier Positional Encodings, Neural Operators.                                                |
| Theoretical Mathematics              | Symplectic Geometry & Differential Forms, Canonical Phase Space Topologies, Contact Hamiltonian Mechanics, Nonlinear Parabolic & Elliptic PDEs, Free-Boundary Obstacle Problems, Optimal Transport & Wasserstein Metrics, Functional Analysis, Operator Theory. |
| Autograd & Deep Learning Engineering | PyTorch (low-level custom autograd engines, <code>torch.func.vmap</code> , forward-mode automatic differentiation, Jacobian-Vector Products (JVP), Hessian-vector contractions), CUDA acceleration, GPU memory profiling, mixed-precision arithmetic.           |

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|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Numerical Analysis & Optimization | Finite Element Method (FEM), Finite Difference Schemes (ADI, Red-Black SOR, Crank-Nicolson), Symplectic Runge-Kutta (RK4), L-BFGS (Strong-Wolfe curvature line search), AdamW, Gradient Surgery (PCGrad, CAGrad), Markov Chain Monte Carlo (MCMC, Metropolis-Hastings). |
| Languages & Developer Toolchain   | Python (NumPy, SciPy, Matplotlib, PyTorch), C++, Git/GitHub, Linux/Bash, LaTeX, VS Code, Kaggle GPU Environments.                                                                                                                                                       |

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## Scholarly Philosophy

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| Open Science & Rigor | Firm conviction in reproducible computational science. Every research framework is accompanied by self-contained automated benchmark runners, zero-redundancy unit testing, full ablation matrices, and permissive open-source licensing (MIT / Apache 2.0). |
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